

HIP JOINT

Regular anatomy notes - type, articular surfaces, capsule, ligaments, relations, supply, movements and applied anatomy

Core idea

The hip joint is the largest ball-and-socket synovial joint. It is formed between the head of the femur and the acetabulum of the hip bone.

It is a weight-bearing joint: it allows flexion, extension, abduction, adduction, medial rotation, lateral rotation and circumduction, but its range is less than the shoulder because stability is the priority.

In revision, remember the hip joint through four pillars: deep acetabulum, strong capsule, spiral ligaments and powerful surrounding muscles.

High-yield facts

Type: ball-and-socket synovial joint.

Articulation: head of femur with lunate surface of acetabulum.

Strongest ligament: iliofemoral ligament / Bigelow ligament.

Common dislocation: posterior dislocation.

Nerve at risk: sciatic nerve.

Why this joint is stable

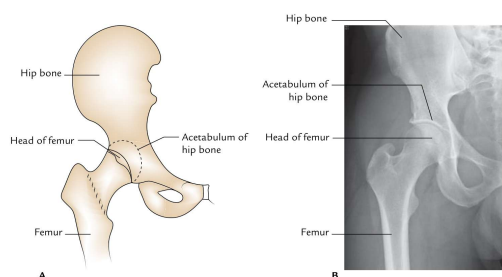
Deep socket: acetabulum surrounds much of femoral head.

Labrum: deepens the acetabular cavity.

Capsule: thick and strong, especially anterosuperiorly.

Ligaments: spiral fibres tighten in extension.

Muscles: gluteal and short rotator muscles add dynamic support.



Source Fig. 26.1 - right hip joint line diagram and radiograph.

General orientation

The head of femur fits into the acetabulum like a ball in a deep cup.

The joint transfers body weight from trunk to femur in standing and walking.

The radiograph helps identify the acetabulum, head of femur and neck-shaft relationship.

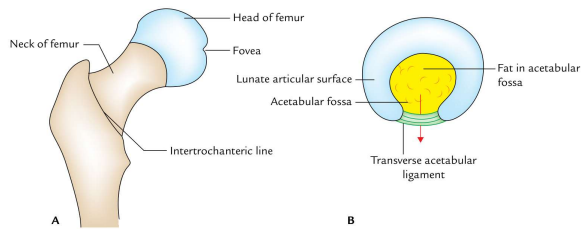
1. Type, functions and basic plan

Feature	Important points
Type	Synovial joint of ball-and-socket variety; it is multiaxial.
Bones involved	Hip bone and femur.
Main functions	Supports body weight during standing and transmits forces generated during walking.
Comparison with shoulder	Both are ball-and-socket joints, but hip is more stable and less mobile; shoulder is more mobile and less stable.
Role of femoral neck	The long and narrow neck of femur acts as a strut and increases range of movement by keeping the shaft away from the pelvis.

Exam starting line

The hip joint is a strong multiaxial ball-and-socket synovial joint between the head of the femur and the acetabulum of the hip bone. It is adapted mainly for weight bearing and stability.

2. Articular surfaces



Source Fig. 26.2 - articular surfaces of hip joint.

How to read the diagram

Head of femur: more than half of a sphere; cartilage covers it except at fovea.

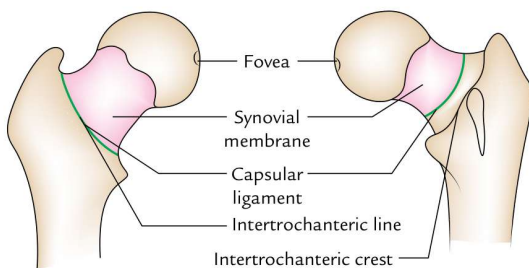
Acetabulum: has lunate surface, acetabular fossa and acetabular notch.

Only lunate surface is articular; acetabular fossa is non-articular and contains fat.

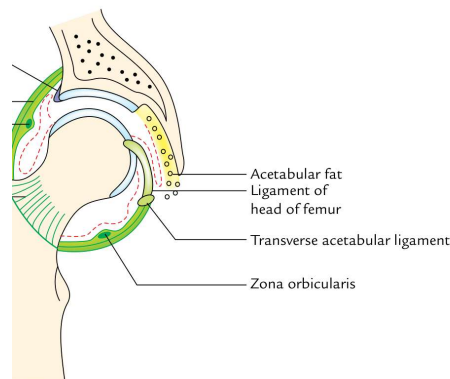
Transverse acetabular ligament bridges the acetabular notch.

Surface / structure	Important details
Head of femur	Forms more than half of a sphere and articulates with the lunate surface of acetabulum. It is covered by hyaline articular cartilage except at fovea capitis.
Fovea capitis	Small pit on head of femur for attachment of ligament of head of femur.
Acetabulum	Cup-shaped cavity of hip bone. It presents lunate surface, acetabular notch and acetabular fossa.
Lunate surface	Horseshoe-shaped articular surface. It is the actual cartilage-covered articular area of acetabulum.
Acetabular fossa	Central non-articular depression containing fat.
Acetabular labrum	Fibrocartilaginous rim attached to acetabular margin; deepens the acetabulum and grips the femoral head.
Transverse acetabular ligament	Bridges acetabular notch and converts it into a foramen for acetabular vessels and nerves.

3. Capsule and synovial membrane



Source Fig. 26.3 - capsular attachment on femur.



Source Fig. 26.4 - capsule and synovial membrane.

Part	Description
Capsule	Strong dense fibrous sac enclosing the joint. It is thick anterosuperiorly and thin posteroinferiorly.
Attachment to hip bone	Attached 5-6 mm beyond acetabular margin; also attached to outer aspect of acetabular labrum and transverse acetabular ligament.
Anterior femoral attachment	Attached to intertrochanteric line.
Posterior femoral attachment	Attached about 1 cm in front of intertrochanteric crest.
Zona orbicularis	Circular fibres in capsule forming a collar around the femoral neck; helps hold femoral head inside acetabulum.
Retinacula	Reflected longitudinal fibres running along femoral neck. They carry important retinacular vessels.
Synovial membrane	Lines inner surface of capsule and covers intracapsular non-articular structures; it does not cover articular cartilage.

Why capsule matters clinically

Retinacular vessels: travel along the femoral neck under capsular retinacula.

Intracapsular fracture: can tear these vessels.

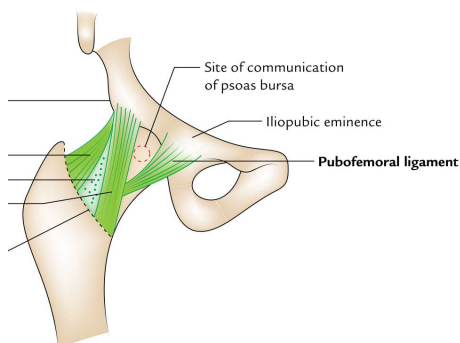
Avascular necrosis: may occur when femoral head loses blood supply.

Posterior weakness

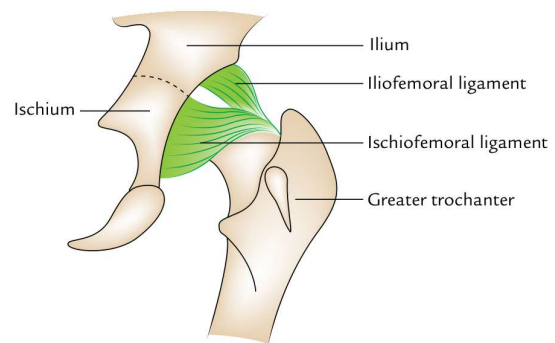
The posteroinferior capsule is relatively weak. In dashboard injury, femoral head may tear this part and dislocate posteriorly. Sciatic nerve lies posteriorly and may be injured.

4. Ligaments of the hip joint

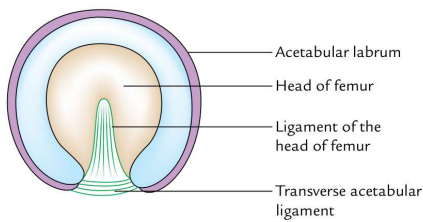
The ligaments of the hip joint reinforce the capsule and check excessive movement. Their spiral arrangement makes them tight during extension and helps maintain erect posture with less muscular effort.



Source Fig. 26.5 - iliofemoral and pubofemoral ligaments.



Source Fig. 26.6 - ischiofemoral and iliofemoral ligaments.



Source Fig. 26.7 - acetabular labrum, transverse acetabular ligament and ligament of head.

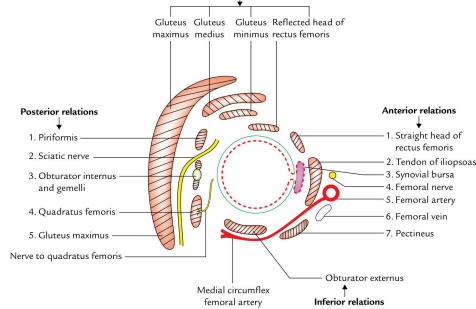
Intra-articular supporting structures

Acetabular labrum increases depth of the socket. Transverse acetabular ligament bridges the acetabular notch.

Ligament of head runs from fovea capitis to transverse acetabular ligament and carries small vessels.

Ligament / structure	Attachments and function
Capsular ligament	Fibrous capsule enclosing the joint.
Iliofemoral ligament	Y-shaped anterior ligament. It extends from AIIS/acetabular margin to intertrochanteric line. It is the strongest ligament of the body and prevents hyperextension.
Pubofemoral ligament	Triangular inferomedial ligament. It checks excessive abduction and extension.
Ischiofemoral ligament	Posterior capsular reinforcement. Fibres spiral from ischial part of acetabular margin to femoral neck/greater trochanter.
Transverse acetabular ligament	Bridges acetabular notch; completes acetabular rim and allows passage of vessels and nerves.
Acetabular labrum	Fibrocartilaginous rim; deepens socket and improves stability.
Ligamentum teres femoris	Ligament of head of femur. It is weak for mechanical stability but transmits small vessels to femoral head.

5. Relations of the hip joint



Source Fig. 26.8 - relations of hip joint.

Relation map

Anterior: iliopsoas tendon, bursa, rectus femoris, femoral nerve, artery and vein.

Posterior: short lateral rotators, sciatic nerve and gluteus maximus.

Superior: reflected head of rectus femoris and gluteal muscles.

Inferior: pectineus and obturator externus.

Aspect	Structures related
Anterior	Iliopsoas tendon separated by synovial bursa; lateral part of pectineus; straight head of rectus femoris; femoral nerve, femoral artery and femoral vein.
Posterior	Piriformis, obturator externus, obturator internus and gemelli, quadratus femoris, gluteus maximus, sciatic nerve, posterior cutaneous nerve of thigh and nerve to quadratus femoris.
Superior	Reflected head of rectus femoris medially; gluteus minimus, medius and maximus laterally.

Aspect	Structures related
Inferior	Pectineus and obturator externus.
Vascular relation	Medial circumflex femoral artery is especially important because its retinacular branches supply most of femoral head and neck.

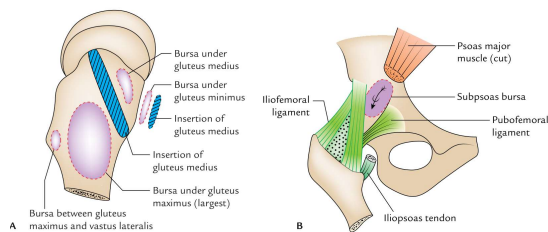
Clinical meaning of relations

Posterior dislocation may injure the sciatic nerve because it lies behind the joint.

The iliopsoas tendon and subpsoas bursa lie anteriorly, so anterior hip pain can be related to iliopsoas irritation.

The medial circumflex femoral artery must be protected during operations around femoral neck.

6. Bursae around the hip joint



Source Fig. 26.9 - bursae around hip joint.

Bursae reduce friction

Seven bursae are described around the hip joint. Four lie under gluteus maximus.

One lies under gluteus medius and one under gluteus minimus.

The subpsoas bursa may communicate with the hip-joint cavity.

Bursa	Important point
Bursa between gluteus maximus and ilium	Lies near the smooth area of ilium between posterior curved line and outer lip of iliac crest.
Trochanteric bursa	Between gluteus maximus and lower outer part of greater trochanter; clinically important in lateral hip pain.
Ischial bursa	Between gluteus maximus and ischial tuberosity; inflammation/enlargement is called weaver bottom.
Bursa between gluteus maximus and vastus lateralis	Reduces friction between muscle/tendon and lateral femoral region.
Bursa under gluteus medius	Between gluteus medius and upper lateral aspect of greater trochanter.
Bursa under gluteus minimus	Between gluteus minimus and anterior aspect of greater trochanter.
Subpsoas bursa	Between iliopubic eminence and psoas tendon; may communicate with hip joint cavity in about 10% of individuals.

Bursitis clinical clues

Lateral hip pain: commonly points toward trochanteric bursa irritation.

Pain on sitting: may involve ischial bursa near ischial tuberosity.

Anterior groin pain: may be related to iliopsoas tendon or subpsoas bursa.

Practical exam link

Show greater trochanter and identify bursae around it.

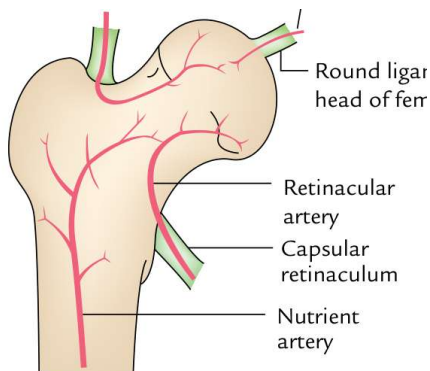
Relate subpsoas bursa to anterior capsule and iliopsoas tendon.

Remember subpsoas bursa may communicate with joint cavity.

Repeated friction: inflames bursae around greater trochanter and gluteal insertions.

Use bursae to explain why extra-articular hip pain can mimic joint pain.

7. Blood supply, nerve supply and segmental control



Source Fig. 26.10 - arterial supply of head of femur.

Most important vessel concept

Retinacular vessels are the key adult supply to femoral head and neck.

They mainly come from the medial circumflex femoral artery.

They run along the femoral neck beneath the capsule.

Intracapsular fractures can tear them and cause avascular necrosis.

Arterial supply

Medial circumflex femoral artery: most important source through retinacular branches.

Lateral circumflex femoral artery: contributes to periarticular arterial network.

Obturator artery: acetabular branch may pass through ligament of head.

Superior/inferior gluteal arteries: contribute around the joint.

Nutrient artery of femur: gives small branches to femoral neck/head.

Nerve supply

Femoral nerve: branch through nerve to rectus femoris.

Obturator nerve: articular twig from anterior division.

Accessory obturator nerve: if present, may supply joint.

Nerve to quadratus femoris: posterior articular supply.

Superior gluteal nerve and sciatic nerve: may contribute.

Segment	Movement group controlled
L2, L3	Flexion, adduction and medial rotation of hip.
L4, L5	Extension, abduction and lateral rotation of hip.

8. Movements and muscles

Movement	Chief muscles / notes
Flexion	Psoas major and iliacus are chief flexors; sartorius, rectus femoris and pectineus assist.
Extension	Gluteus maximus is chief extensor; hamstrings assist.
Abduction	Gluteus medius and gluteus minimus are chief abductors; tensor fasciae latae and sartorius assist.
Adduction	Adductor longus, adductor brevis and adductor magnus are chief adductors; pectineus and gracilis assist.
Medial rotation	Anterior fibres of gluteus medius and minimus are chief medial rotators;

Movement	Chief muscles / notes
	tensor fasciae latae assists.
Lateral rotation	Piriformis, obturator internus, obturator externus, superior and inferior gemelli and quadratus femoris are short lateral rotators.
Circumduction	Combination of flexion, extension, abduction and adduction.

Range of movement

Flexion: about 110-120 degrees; limited by thigh touching abdomen.

Extension: about 15 degrees.

Abduction: about 50 degrees.

Adduction: limited by contact with opposite thigh.

Medial rotation: about 25 degrees.

Lateral rotation: about 60 degrees.

Axis memory

Transverse axis: flexion and extension.

Anteroposterior axis: abduction and adduction.

Vertical axis: medial and lateral rotation.

Combined axes: circumduction.

Trendelenburg concept

Gluteus medius and minimus keep the pelvis level when standing on one limb.

Weak abductors or superior gluteal nerve lesion causes pelvic drop on the opposite side.

This produces positive Trendelenburg sign and lurching gait.

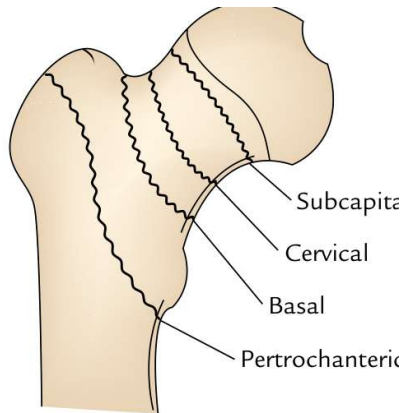
9. Stability of the hip joint

Stabilizing factor	How it works
Depth of acetabulum	Deep cup prevents easy displacement of femoral head.
Acetabular labrum	Narrows and deepens acetabular mouth, increasing grip on head of femur.
Strong capsule	Dense fibrous capsule encloses and supports the joint.
Iliofemoral ligament	Checks hyperextension and supports standing posture.
Pubofemoral and ischiofemoral ligaments	Limit excessive abduction/extension and reinforce inferior/posterior capsule.
Muscle tone	Gluteal muscles and short rotators act as dynamic stabilizers.
Atmospheric pressure / suction	Helps keep articular surfaces opposed.

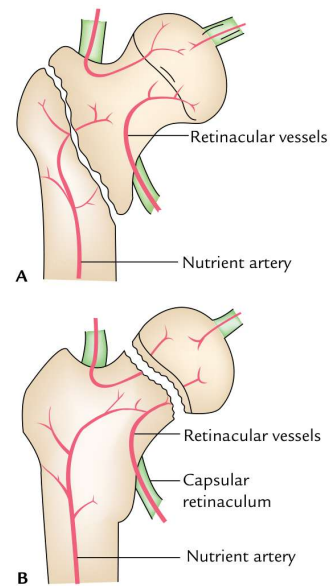
Revision comparison

Shoulder joint gives maximum range but sacrifices stability. Hip joint sacrifices some range to provide high stability for weight bearing.

10. Applied anatomy



11 Types of fracture of the neck of femur.
Source Fig. 26.11 - types of fracture of neck of femur.



Source Fig. 26.12 - vessel involvement in femoral neck fractures.

Applied point	Explanation
Congenital dislocation	More common than congenital dislocation of other joints. It may be due to loose capsule at birth and hypoplastic acetabulum or femoral head. Signs include limited abduction, limb shortening, external rotation, asymmetrical thigh folds, lurching gait and positive Trendelenburg sign.
Acquired dislocation	Uncommon because the hip is strong and stable. It classically occurs in dashboard injury when hip is flexed, adducted and medially rotated.
Posterior dislocation	Most common type. Femoral head tears posteroinferior capsule and lies on lateral surface of ilium. Limb becomes shortened and medially rotated. Sciatic nerve may be injured.
Perthes disease	Destruction and flattening of femoral head with increased joint space on radiograph.
Coxa vara	Neck-shaft angle is reduced. It can occur in fracture neck of femur, Perthes disease or rickets.
Coxa valga	Neck-shaft angle is increased. It may be associated with congenital dislocation of hip.
Osteoarthritis	Degenerative disease of old age. Osteophytes at articular ends limit movement and make movements grating and painful.
Referred pain to knee	Hip disease may present as knee pain because hip and knee share part of their nerve supply.
Aspiration of hip joint	Needle is inserted about 5 cm below ASIS and directed upward, backward and medially.

11. Fracture neck of femur

Fracture neck of femur is common in elderly individuals, especially females, because the femoral neck becomes weak and brittle due to osteoporosis. It is often incorrectly called a fractured hip.

Fracture type	Key point
Subcapital	Near the head of femur; intracapsular and dangerous for retinacular vessels.
Cervical	In the middle of femoral neck; intracapsular.
Basal	Near the trochanters.
Pretrochanteric / pertrochanteric	Just distal to trochanters; extracapsular, so retinacular vessels are usually spared.

Clinical position after fracture

The patient falls and cannot get up.
Affected limb is shortened.
Affected limb is laterally rotated.
Diagnosis is confirmed by X-ray.

Radiological lines

Shenton line: curved line formed by upper border of obturator foramen and lower margin of femoral neck; broken in fracture/dislocation.

Schoemaker line: line from greater trochanter to ASIS and upward to abdomen; if greater trochanter is elevated, line passes below umbilicus.

Avascular necrosis trap

Retinacular vessels from medial circumflex femoral artery supply most of head and neck of femur. Intracapsular fractures such as subcapital and cervical fractures may damage them. Pretrochanteric extracapsular fracture usually does not damage them, so avascular necrosis is less likely.

12. Final revision page

Question	Answer
What type of joint is hip joint?	Ball-and-socket synovial joint.
What are the articular surfaces?	Head of femur and lunate surface of acetabulum.
Which structure deepens acetabulum?	Acetabular labrum.
What is the strongest ligament?	Iliofemoral ligament / ligament of Bigelow.
Which ligament is Y-shaped?	Iliofemoral ligament.
Which ligament attaches to fovea capitis?	Ligamentum teres femoris / ligament of head of femur.
Main blood supply of femoral head in adults?	Retinacular vessels mainly from medial circumflex femoral artery.
Commonest dislocation?	Posterior dislocation.
Nerve injured in posterior dislocation?	Sciatic nerve.
Common clinical fracture?	Fracture neck of femur.
Why can hip disease cause knee pain?	Because of common nerve supply of hip and knee joints.
What is weaver bottom?	Enlarged/inflamed ischial bursa between gluteus maximus and ischial tuberosity.
Normal neck-shaft angle?	About 120 degrees in adults and 160 degrees in children.

Do not forget in practicals

Identify acetabulum, head, neck and greater trochanter on model/X-ray.
Trace iliofemoral ligament anteriorly.
Relate sciatic nerve posterior to the joint.
Remember bursa under psoas may communicate with joint cavity.

One-line summary

Hip joint is a stable ball-and-socket joint formed by head of femur and acetabulum.
Its stability depends on acetabular depth, labrum, strong capsule, ligaments and muscles.
Posterior dislocation and femoral-neck fracture are the key applied anatomy topics.

Exam traps

Separate articular surfaces from ligaments. Do not call the whole acetabulum articular - only the lunate surface is articular. Do not forget medial circumflex femoral artery, posterior dislocation, sciatic nerve injury and referred knee pain.

13. Exam writing checklist

Heading in answer	What to write under it
Definition	Largest ball-and-socket synovial joint between head of femur and acetabulum of hip bone.
Articular surfaces	Head of femur and lunate surface of acetabulum. Mention fovea capitis, acetabular fossa, acetabular notch and labrum.
Capsule	Strong fibrous capsule; thick anterosuperiorly and thin posteroinferiorly. Mention attachments to acetabular margin, intertrochanteric line and neck of femur.
Ligaments	Write capsular ligament, iliofemoral, pubofemoral, ischiofemoral, transverse acetabular ligament, acetabular labrum and ligamentum teres.
Relations	Divide into anterior, posterior, superior and inferior relations. Mention sciatic nerve posteriorly and medial circumflex femoral artery.
Supply	Blood supply from circumflex femoral, obturator and gluteal arteries; nerve supply from femoral, obturator, nerve to quadratus femoris and superior gluteal nerves.
Movements	Write flexion, extension, abduction, adduction, medial rotation, lateral rotation and circumduction with chief muscles.
Applied anatomy	Posterior dislocation, fracture neck of femur, avascular necrosis, coxa vara/valga, osteoarthritis and referred pain to knee.

Diagram labelling checklist

Fig. 26.2: label head of femur, fovea, lunate surface, acetabular fossa and transverse acetabular ligament.

Fig. 26.5: label iliofemoral ligament and pubofemoral ligament.

Fig. 26.6: label ischium, ilium, iliofemoral ligament and ischiofemoral ligament.

Clinical case clues

Dashboard injury: think posterior dislocation of hip.

Shortened medially rotated limb: posterior dislocation.

Shortened laterally rotated limb in elderly: fracture neck of femur.

Knee pain with hip disease: referred pain from

Fig. 26.8: mark anterior and posterior relations separately.

common nerve supply.

Last-minute memory hooks

Articular surface = head of femur + lunate surface only.

Strongest ligament = iliofemoral ligament.

Main adult vessel = medial circumflex femoral artery via retinacular branches.

Commonest dislocation = posterior; nerve at risk = sciatic nerve.

Intracapsular femoral-neck fracture = risk of avascular necrosis.

Gluteus medius/minimus weakness = Trendelenburg sign.

14. Self-test before closing the topic

Prompt	Expected answer
Name two structures increasing stability.	Acetabular labrum and iliofemoral ligament; also mention deep acetabulum and muscle tone.
Which part of acetabulum is articular?	The lunate surface only.
Why is hip less mobile than shoulder?	Because the acetabulum is deeper and the capsuloligamentous support is much stronger for weight bearing.
Which ligament prevents hyperextension?	Iliofoemoral ligament.
Which vessel is most important in adults?	Medial circumflex femoral artery through retinacular vessels.
What happens in posterior dislocation?	Femoral head tears posteroinferior capsule; limb is shortened and medially rotated; sciatic nerve may be injured.